

Linear Regression Model for Council District VI

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City of Detroit

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Contents

Model I	1
Model II	5
Model III	9
Model IV	17

Model I

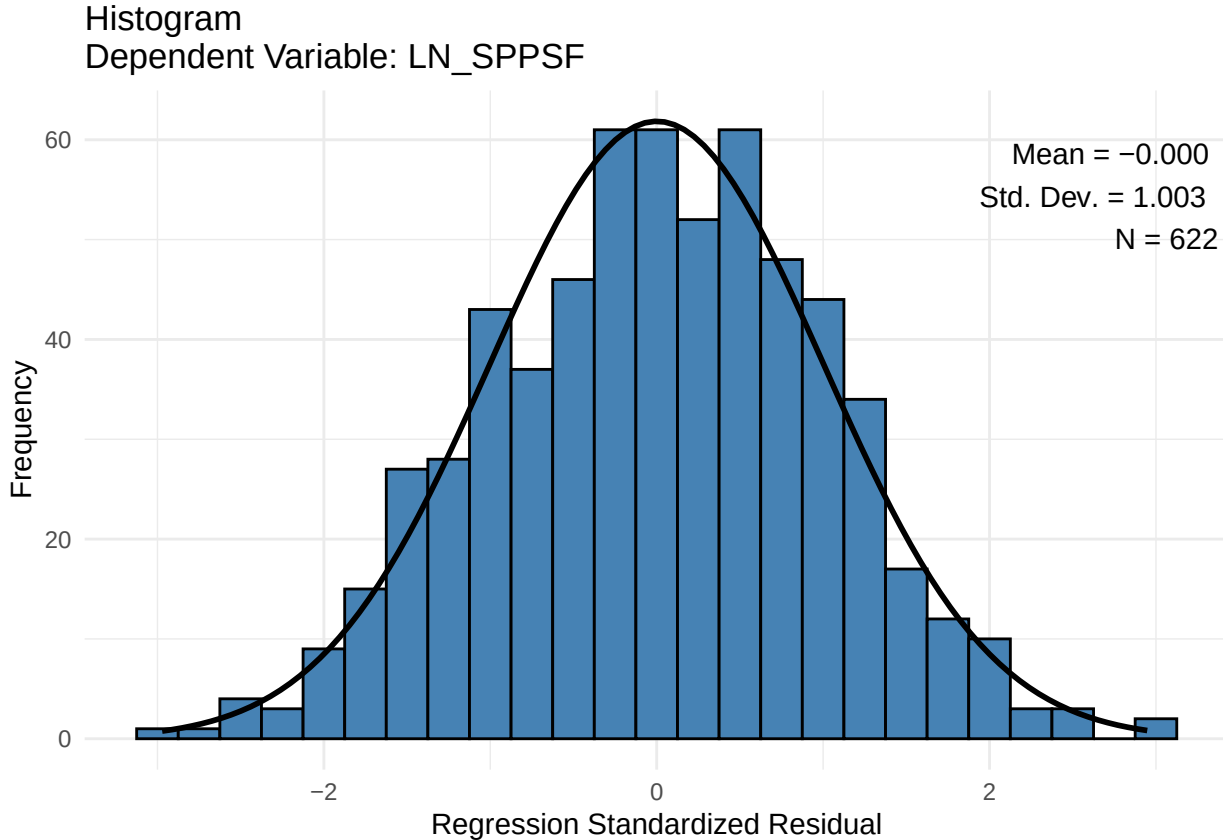


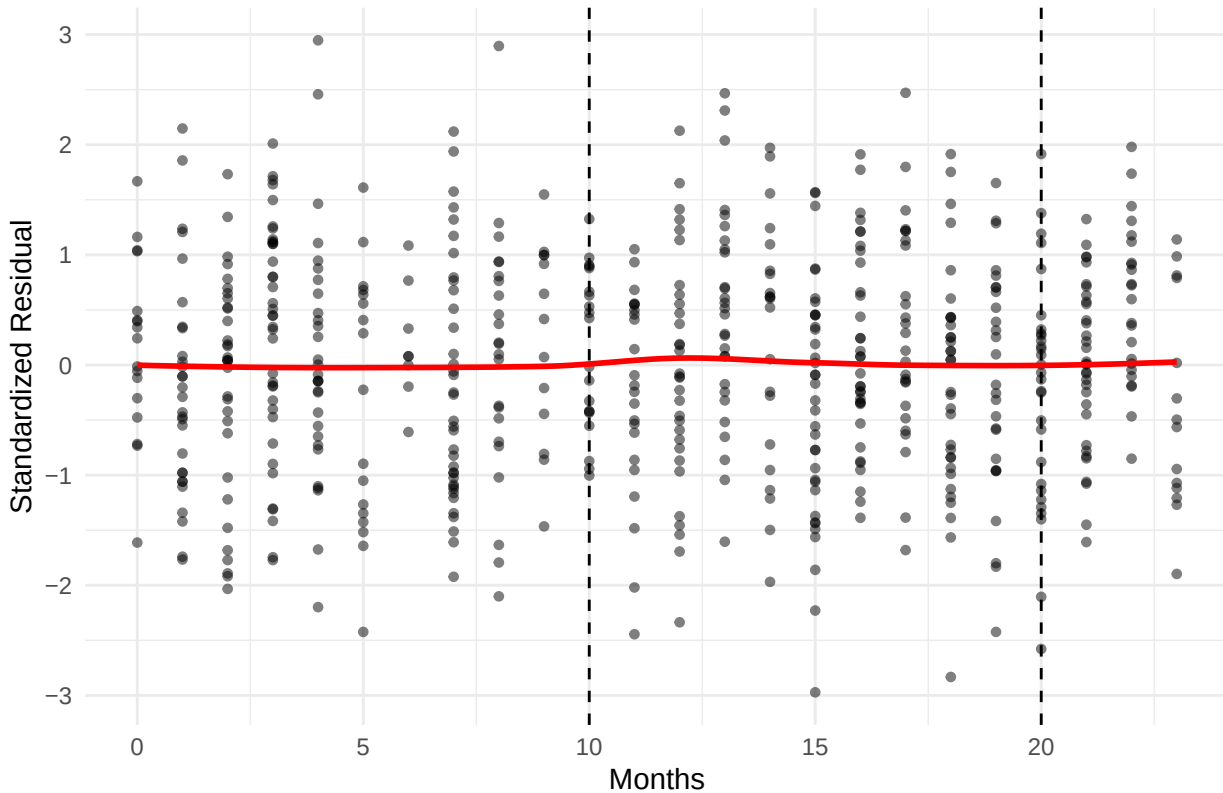
Table 1: District 6 LN_SPPSF model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
622	608	16	0.5922	0.2377	0.2214	0.5855	0.4742	14.5806	0	1129.176	1195.67

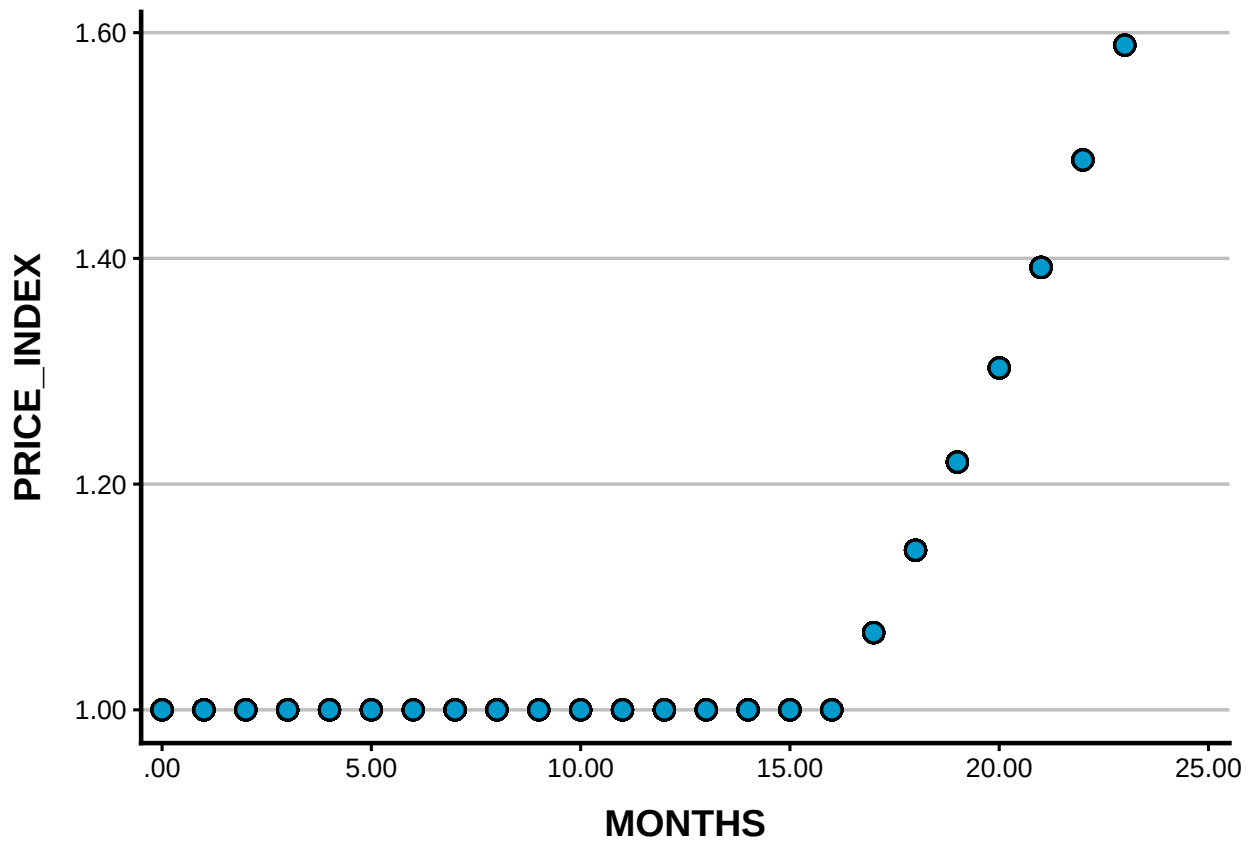
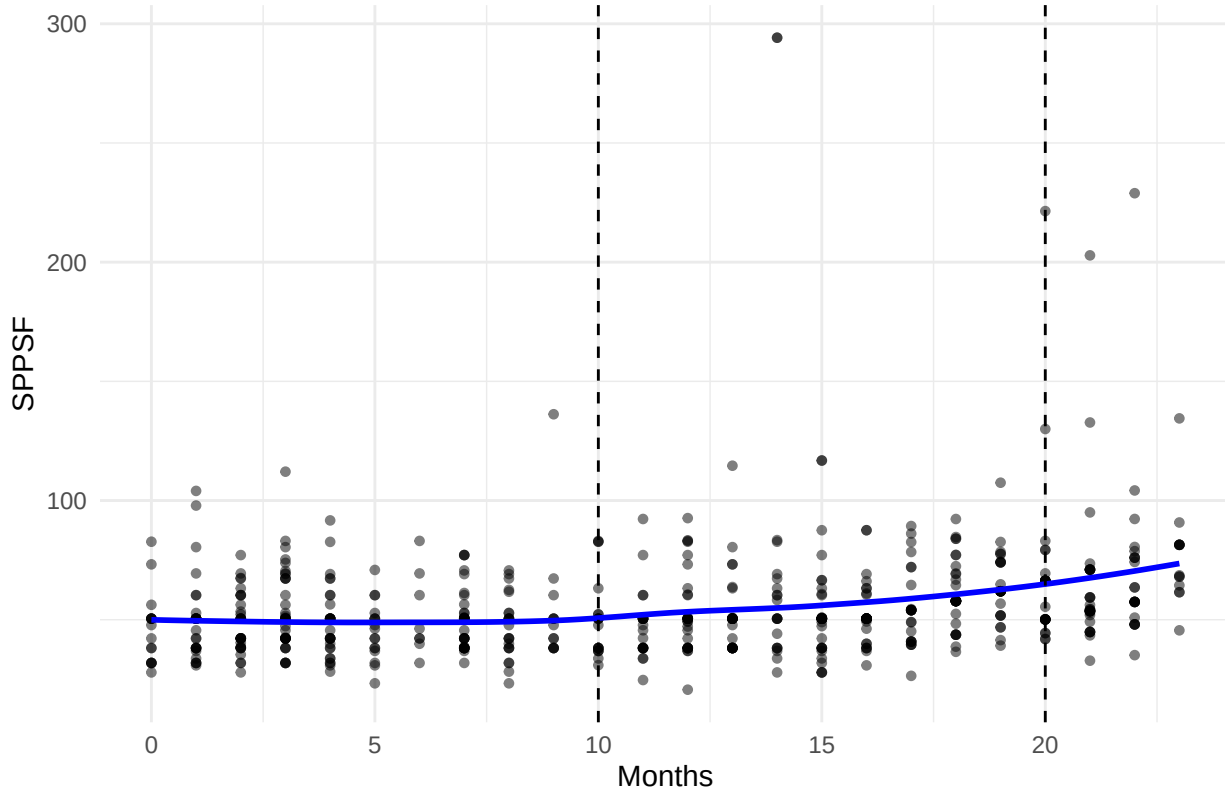
Table 2: District 6 LN_SPPSF model coefficients (95% CI)

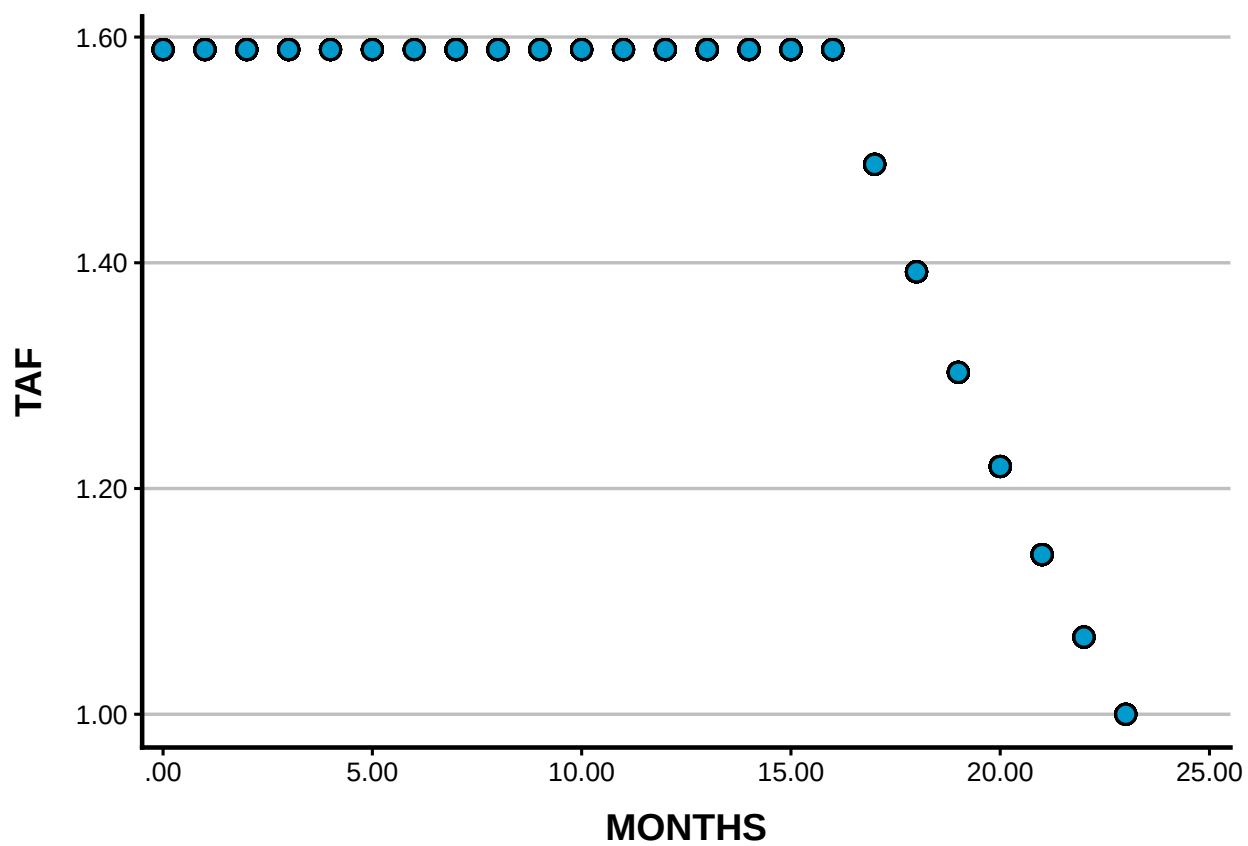
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	3.7417	65.5581	0.0571	0.0000	3.6296	3.8538
arterial	0.2882	3.5514	0.0812	0.0004	0.1288	0.4476
single_family1story	0.1792	3.2370	0.0554	0.0013	0.0705	0.2880
half_duplex1story	NA	NA	NA	NA	NA	NA
largest	0.2694	2.3511	0.1146	0.0190	0.0444	0.4943
two_plus_baths	0.3343	2.1774	0.1535	0.0298	0.0328	0.6358
powder_room	0.2252	2.7221	0.0827	0.0067	0.0627	0.3877
good	0.4986	3.4225	0.1457	0.0007	0.2125	0.7847
fair	-0.2801	-5.3460	0.0524	0.0000	-0.3829	-0.1772
poor	-0.4016	-3.6022	0.1115	0.0003	-0.6205	-0.1826
qabove_avg	0.6170	2.0314	0.3037	0.0426	0.0205	1.2135
yb_post_1961	0.3727	2.9673	0.1256	0.0031	0.1260	0.6193
fireplace	-0.3123	-3.9977	0.0781	0.0001	-0.4657	-0.1589
elec_baseboard	NA	NA	NA	NA	NA	NA
central_air	0.1785	2.0676	0.0863	0.0391	0.0090	0.3480
months_17to24	0.0684	5.6556	0.0121	0.0000	0.0446	0.0921

Residuals for LN_SPPSF Model



Months vs Predicted SPPSF – LN_SPPSF Model





Model II

Histogram
Dependent Variable: LN_PRICE (No ECFs)

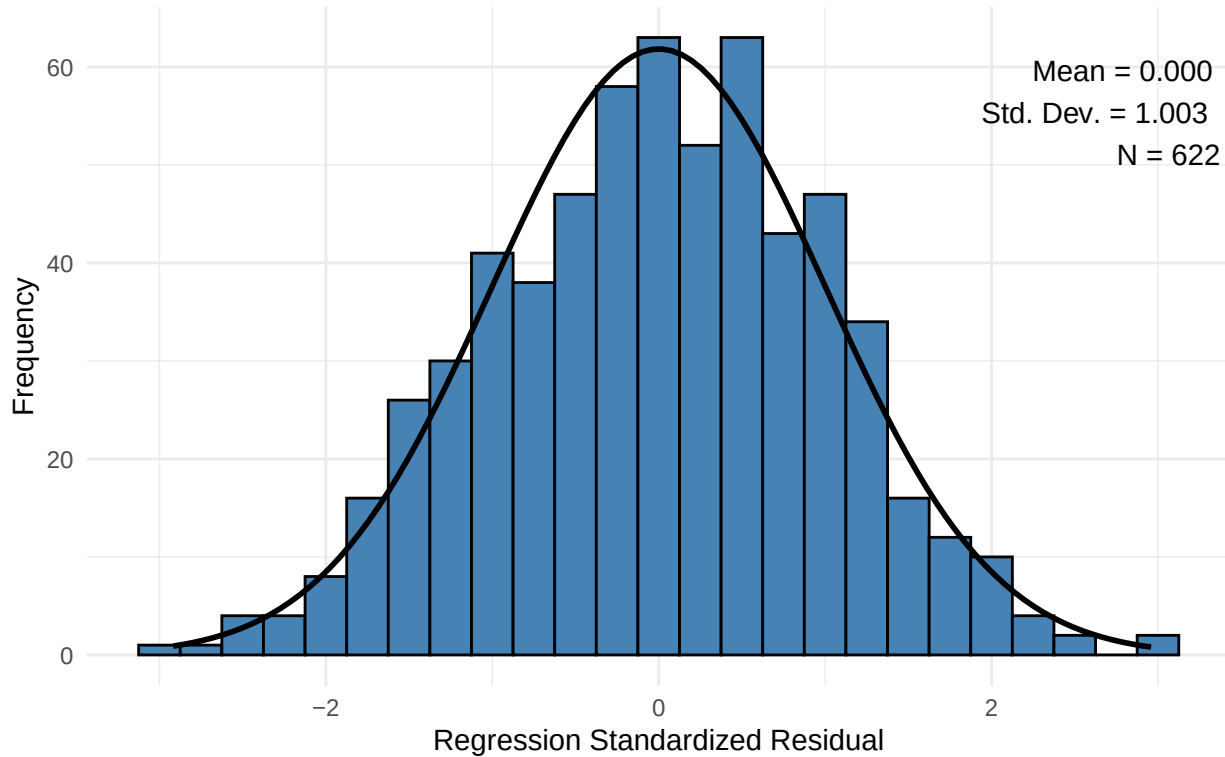


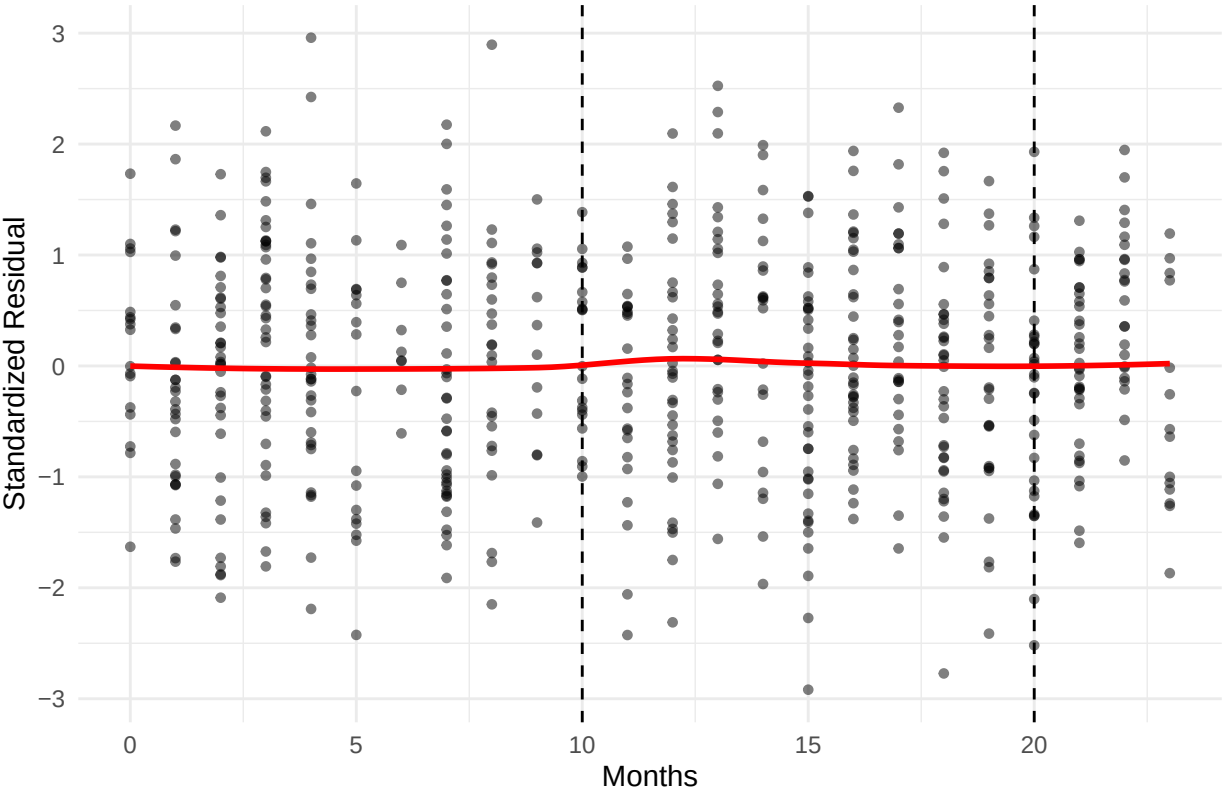
Table 3: District 6 LN_PRICE (No ECFs) model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
622	607	17	0.5922	0.3983	0.3845	0.5851	0.4739	28.7039	0	1130.318	1201.245

Table 4: District 6 LN_PRICE (No ECFs) model coefficients (95% CI)

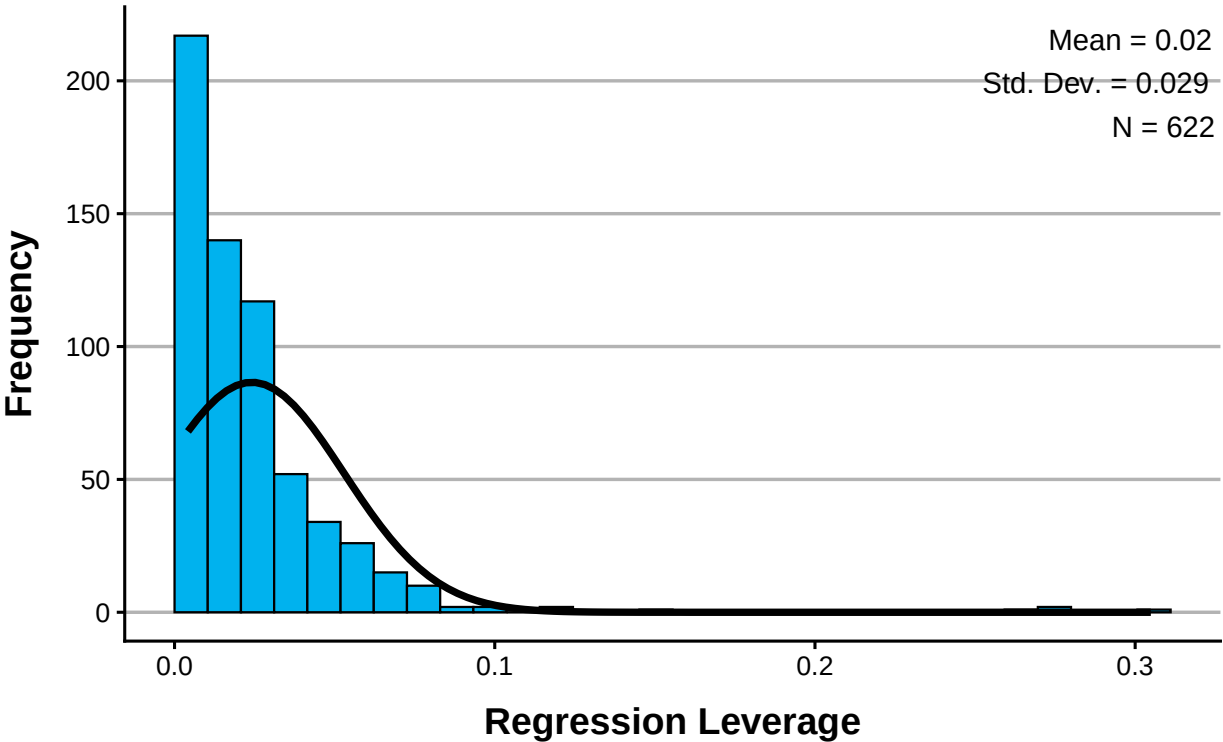
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.7193	179.8778	0.0596	0.0000	10.6022	10.8363
ln_base_bldgsize_ratio	0.8668	6.1504	0.1409	0.0000	0.5900	1.1436
arterial	0.2905	3.5784	0.0812	0.0004	0.1311	0.4500
single_family1story	0.1567	2.5966	0.0603	0.0096	0.0382	0.2752
half_duplex1story	NA	NA	NA	NA	NA	NA
largest	0.2993	2.5454	0.1176	0.0112	0.0684	0.5302
two_plus_baths	0.3822	2.3540	0.1624	0.0189	0.0633	0.7010
powder_room	0.2585	2.8313	0.0913	0.0048	0.0792	0.4377
good	0.5053	3.4637	0.1459	0.0006	0.2188	0.7918
fair	-0.2775	-5.2928	0.0524	0.0000	-0.3804	-0.1745
poor	-0.3977	-3.5647	0.1116	0.0004	-0.6168	-0.1786
qabove_avg	0.5571	1.8339	0.3038	0.0672	-0.0395	1.1537
yb_post_1961	0.3841	3.0433	0.1262	0.0024	0.1363	0.6320
fireplace	-0.2983	-3.7657	0.0792	0.0002	-0.4538	-0.1427
elec_baseboard	NA	NA	NA	NA	NA	NA
central_air	0.1743	2.0133	0.0866	0.0445	0.0043	0.3442
months_17to24	0.0684	5.6562	0.0121	0.0000	0.0447	0.0922

Residuals for LN_PRICE (No ECFs) Model

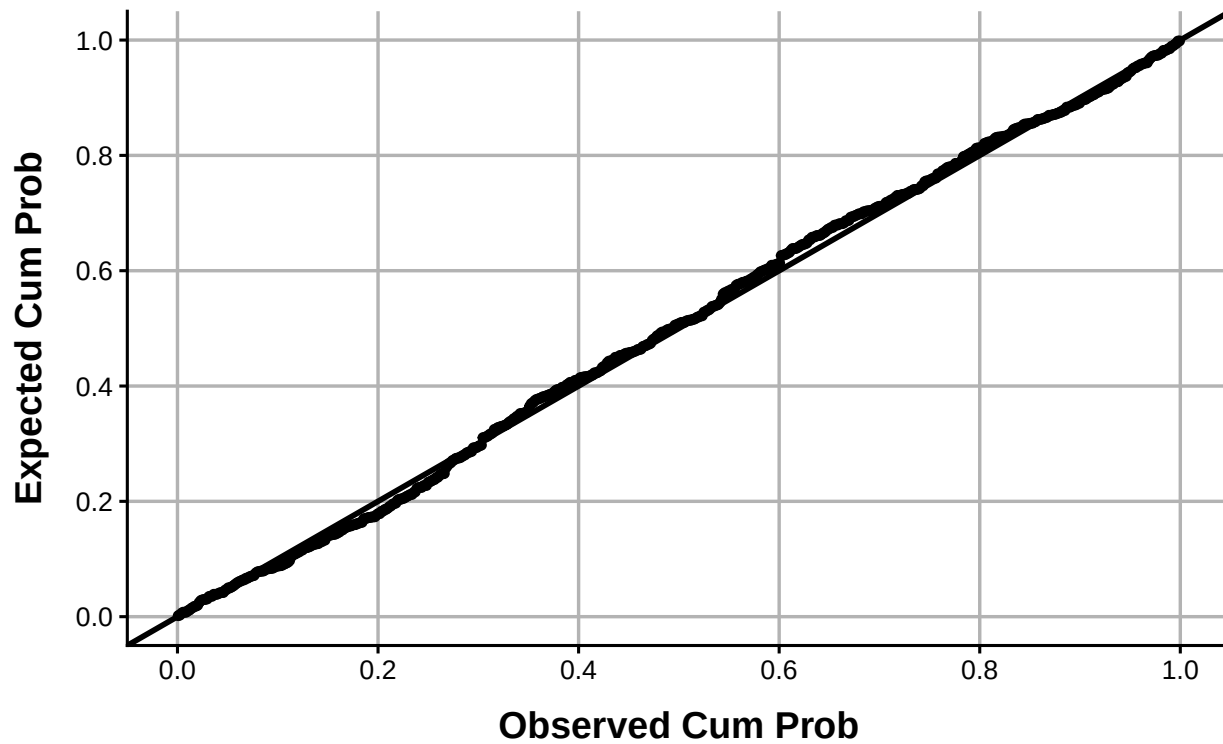


Histogram

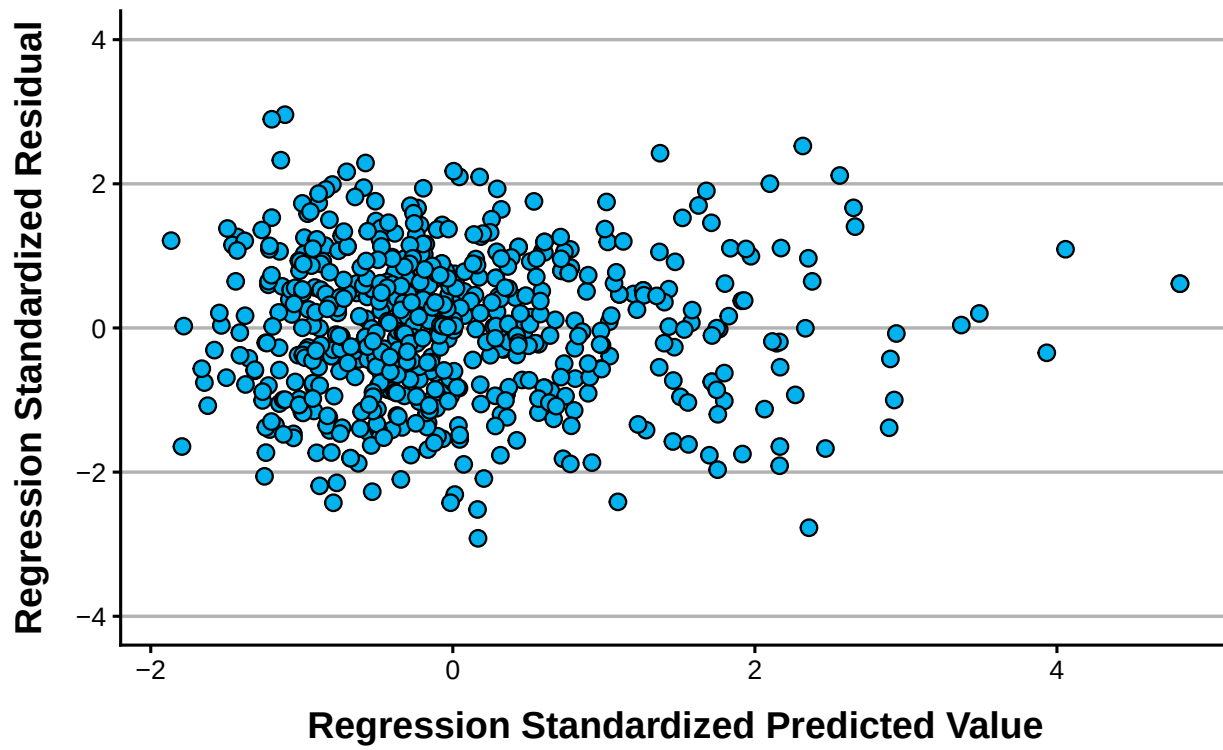
Dependent Variable: LN\ PRICE

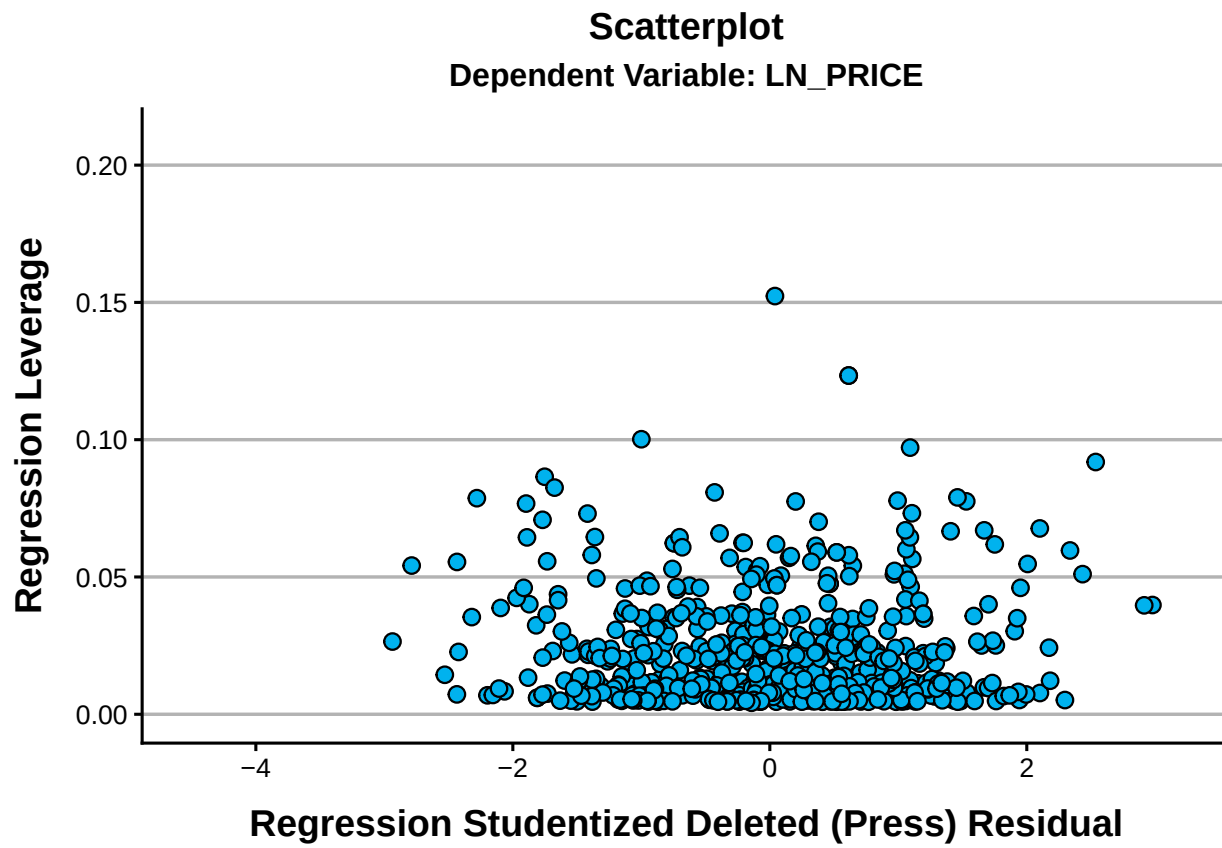
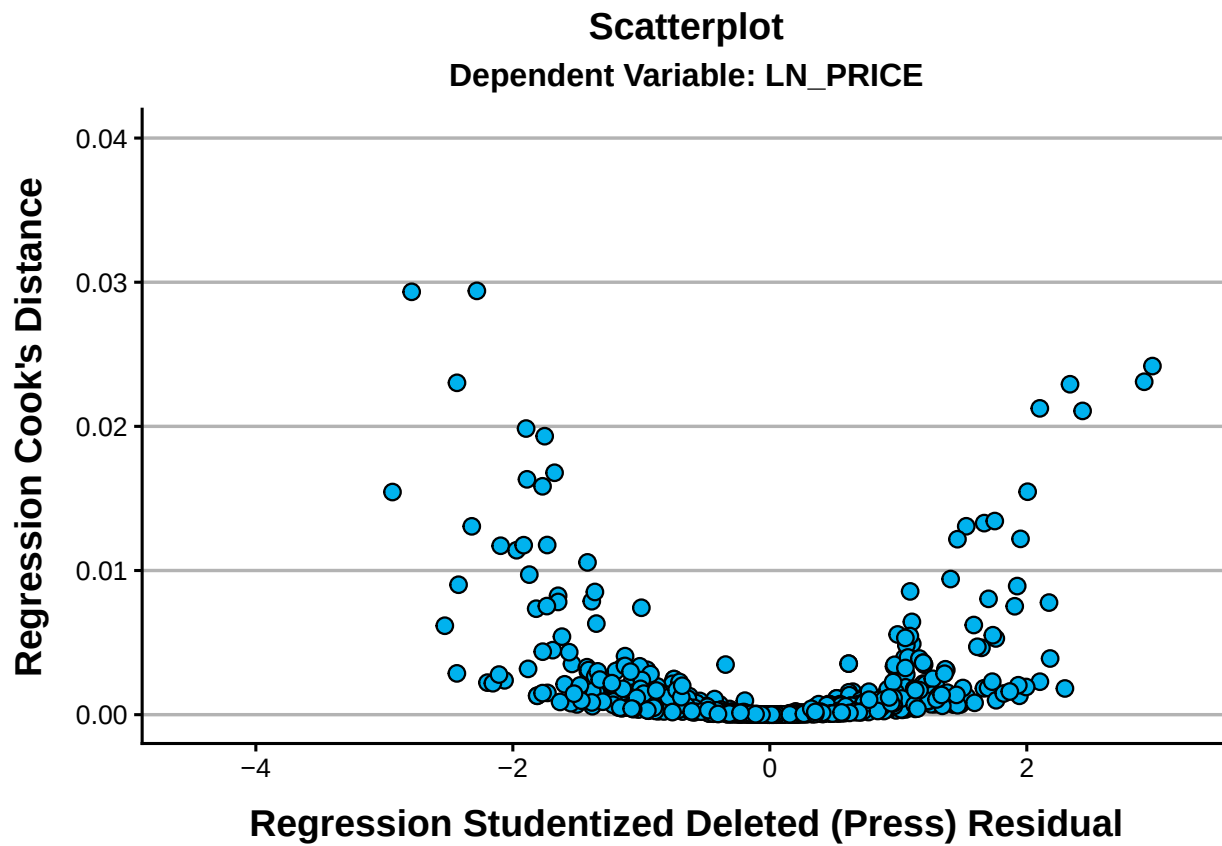


Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE





Model III

Initial Input

Table 5: LN_PRICE model (base) assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
622	0.9598829	1.139136	0.8916394	1.277575	47.4691	-0.5428275

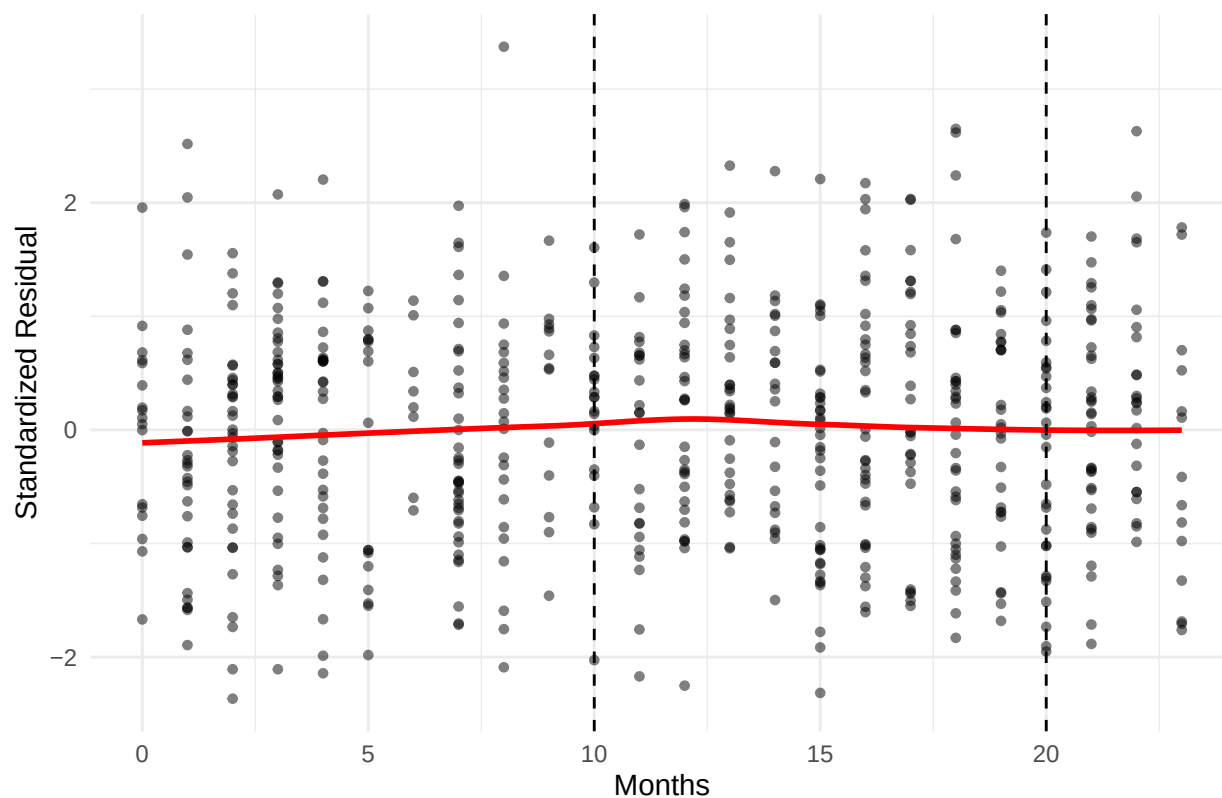
Table 6: District 6 LN_PRICE model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
622	592	32	0.529	0.5318	0.5089	0.5161	0.4215	23.1866	0	1004.312	1141.733

Table 7: District 6 LN_PRICE model coefficients (95% CI)

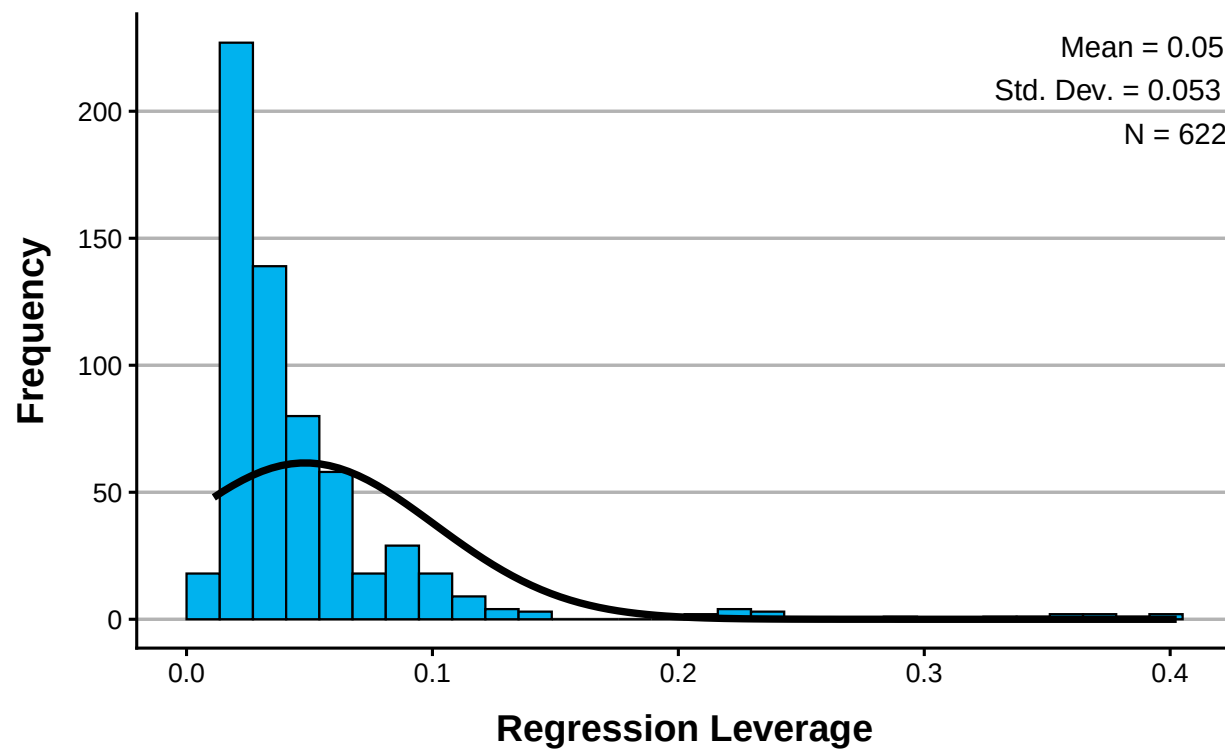
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.4245	134.6152	0.0774	0.0000	10.2724	10.5766
arterial	0.1971	2.6324	0.0749	0.0087	0.0501	0.3442
single_family1story	0.1154	2.1151	0.0545	0.0348	0.0082	0.2225
half_duplex1story	NA	NA	NA	NA	NA	NA
smallest	-0.3424	-4.1438	0.0826	0.0000	-0.5046	-0.1801
small	-0.1540	-2.4492	0.0629	0.0146	-0.2774	-0.0305
large	0.1117	1.7368	0.0643	0.0829	-0.0146	0.2379
largest	0.4292	4.0917	0.1049	0.0000	0.2232	0.6351
powder_room	0.1605	2.0821	0.0771	0.0378	0.0091	0.3119
good	0.3364	2.3343	0.1441	0.0199	0.0534	0.6194
fair	-0.2549	-5.3277	0.0478	0.0000	-0.3489	-0.1609
poor	-0.2945	-2.9188	0.1009	0.0036	-0.4927	-0.0963
qabove_avg	0.5736	2.0129	0.2850	0.0446	0.0139	1.1334
yb_post_1961	0.3467	2.9278	0.1184	0.0035	0.1141	0.5793
ln_garagespaces	0.0907	1.7946	0.0505	0.0732	-0.0086	0.1899
elec_baseboard	NA	NA	NA	NA	NA	NA
ecf_6R604	0.3126	3.9264	0.0796	0.0001	0.1562	0.4689
ecf_6R605	0.5097	5.2969	0.0962	0.0000	0.3207	0.6987
ecf_6R606	0.3006	2.2370	0.1344	0.0257	0.0367	0.5644
ecf_6R607	0.8576	6.3405	0.1353	0.0000	0.5919	1.1232
ecf_6R608	1.7360	8.7199	0.1991	0.0000	1.3450	2.1269
ecf_6R609	2.0219	6.2590	0.3230	0.0000	1.3875	2.6564
ecf_6R611	0.6098	8.1047	0.0752	0.0000	0.4621	0.7576
ecf_6R612	0.5983	4.6645	0.1283	0.0000	0.3464	0.8502
ecf_6R613	0.8499	7.8282	0.1086	0.0000	0.6367	1.0631
ecf_6R614	1.5268	5.9814	0.2553	0.0000	1.0255	2.0281
ecf_6R615	0.6317	6.3677	0.0992	0.0000	0.4369	0.8266
ecf_6R616	0.8124	3.2498	0.2500	0.0012	0.3214	1.3034
ecf_6R617	0.9822	3.1071	0.3161	0.0020	0.3613	1.6030
ecf_6R618	0.3057	1.8146	0.1685	0.0701	-0.0252	0.6366
ecf_6R620	0.2889	3.9636	0.0729	0.0001	0.1457	0.4320
months_17to24	0.0597	5.4576	0.0109	0.0000	0.0382	0.0812

Residuals for LN_PRICE Model

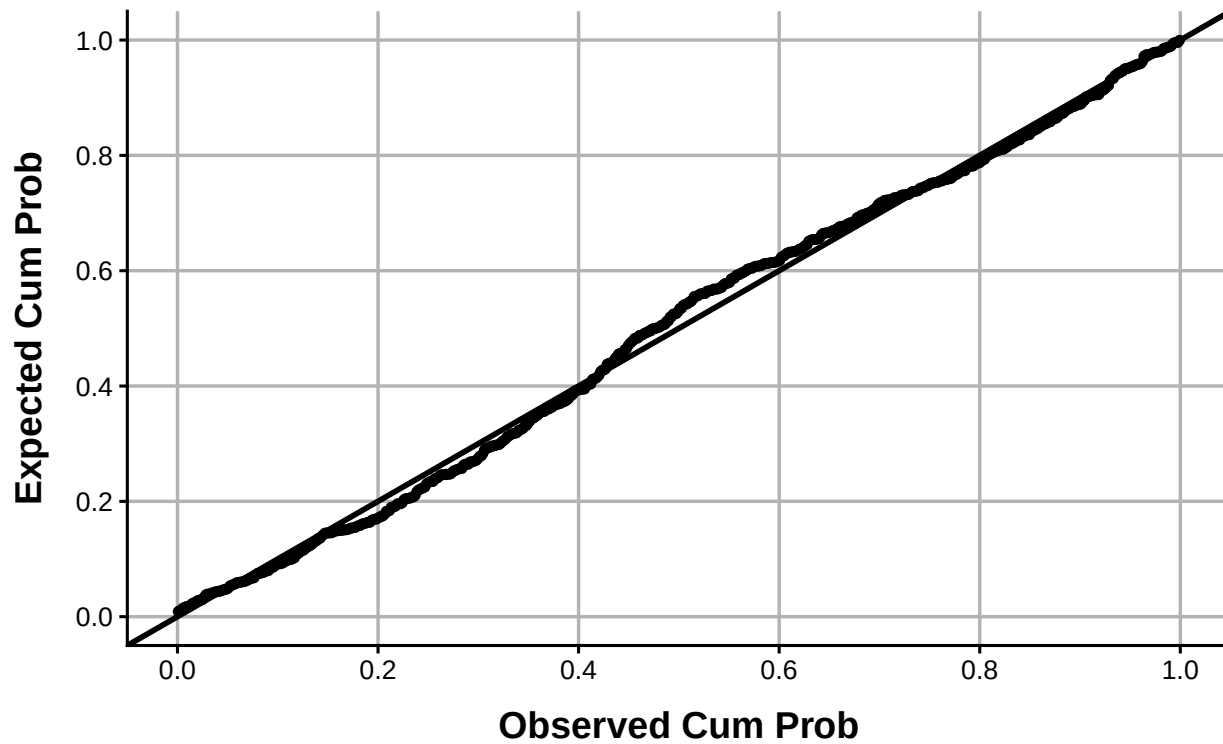


Histogram

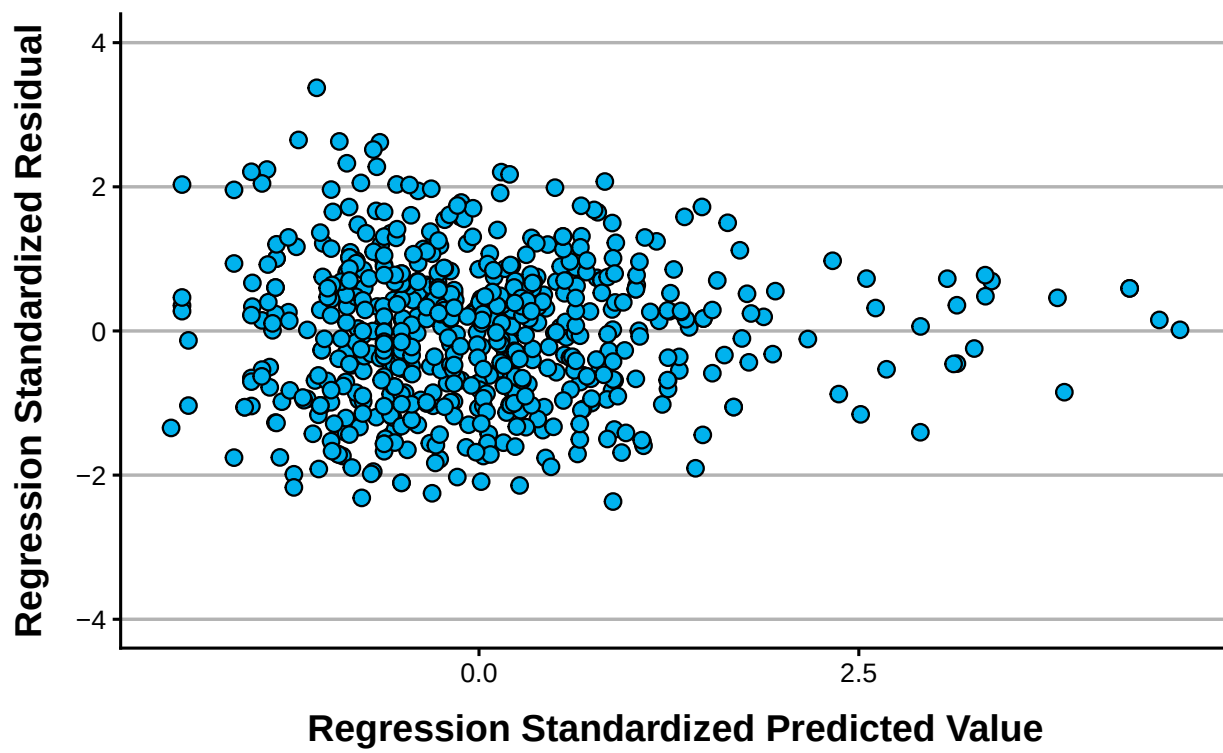
Dependent Variable: LN\ PRICE

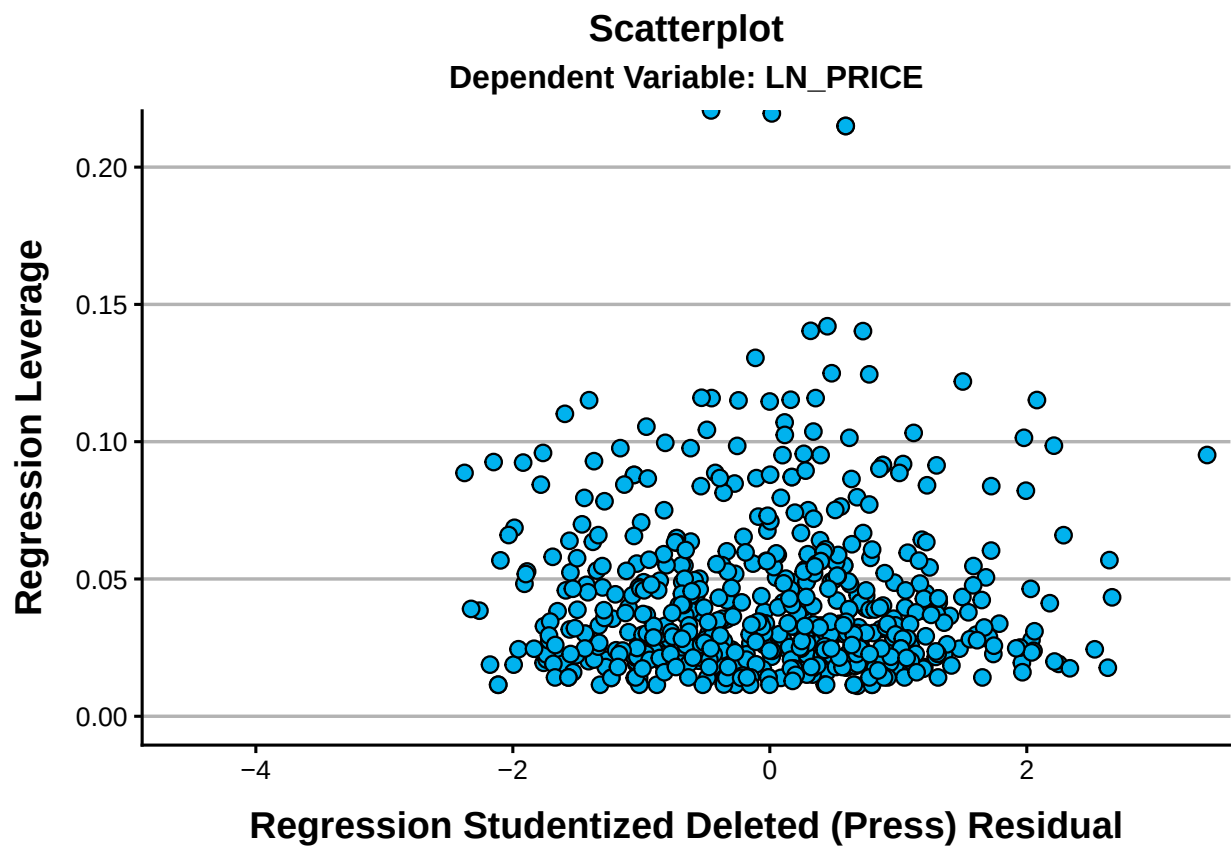
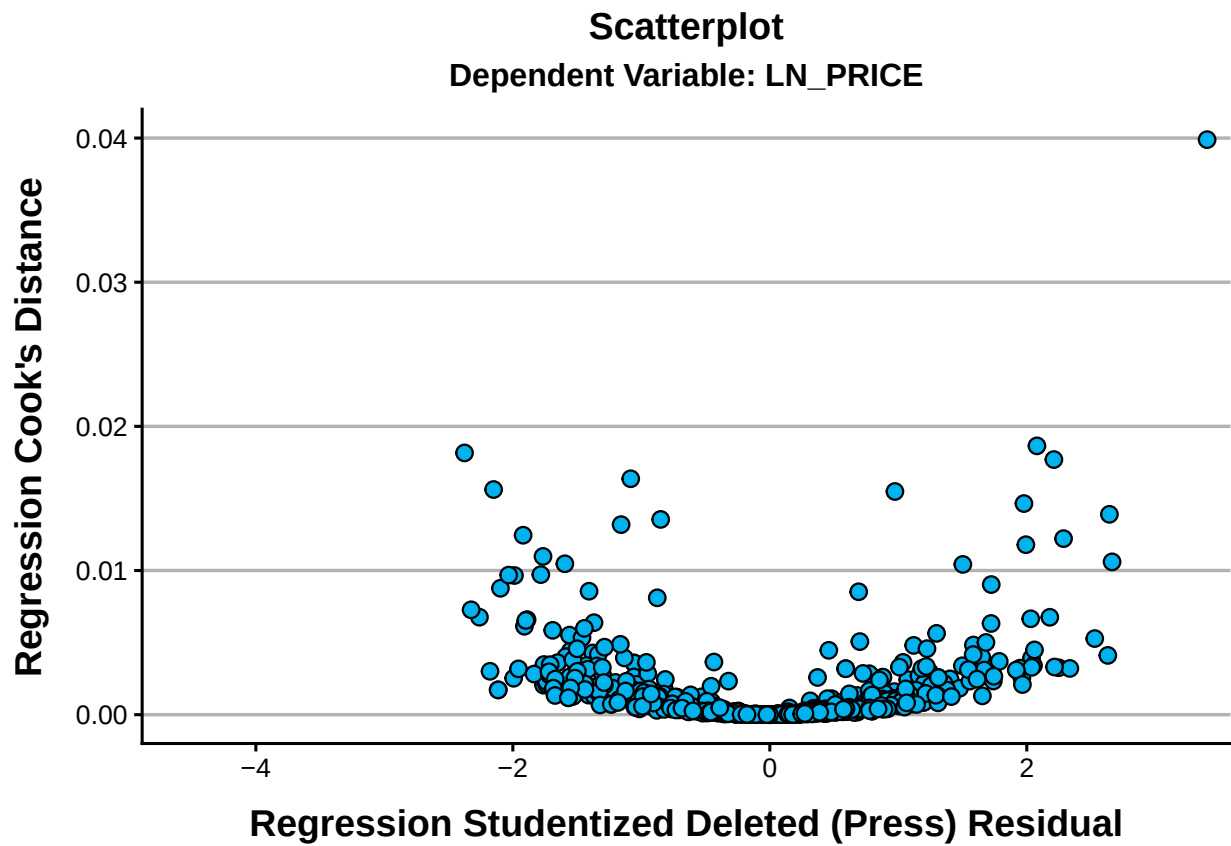


Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE





Filtered Input

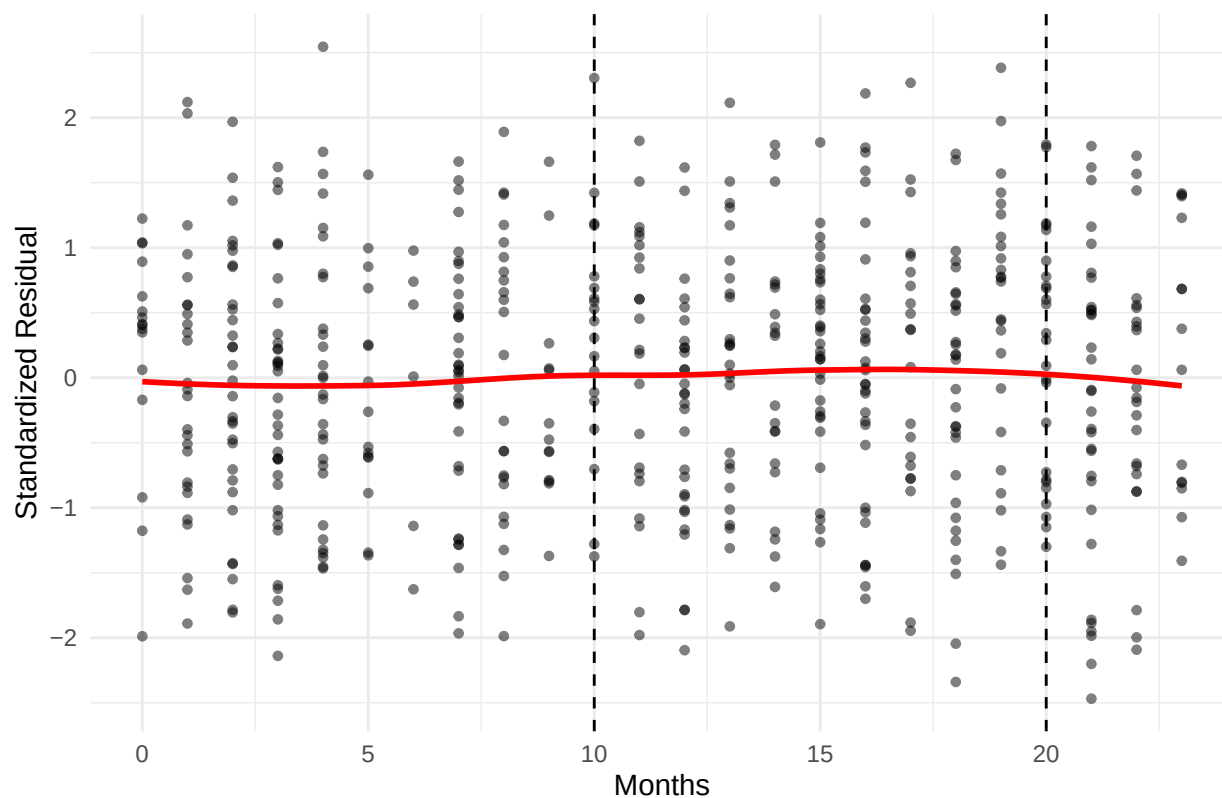
Table 8: District 6 LN_PRICE (Inliers) model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
578	546	34	0.4586	0.6127	0.5907	0.4457	0.3672	27.8621	0	772.1206	915.9865

Table 9: District 6 LN_PRICE (Inliers) model coefficients (95% CI)

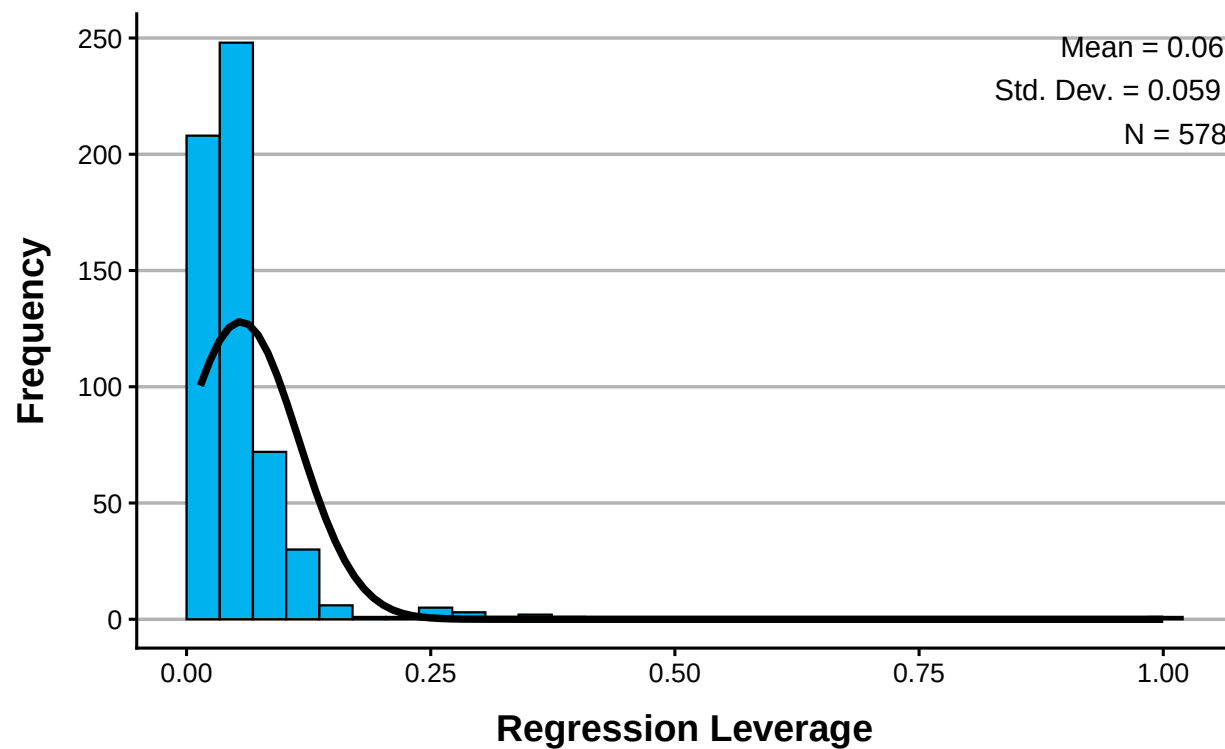
term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.1772	109.6224	0.0928	0.0000	9.9948	10.3596
arterial	0.2534	3.5371	0.0717	0.0004	0.1127	0.3942
single_family1story	0.1396	2.6073	0.0535	0.0094	0.0344	0.2447
half_duplex1story	NA	NA	NA	NA	NA	NA
smallest	-0.3155	-4.1857	0.0754	0.0000	-0.4635	-0.1674
small	-0.1143	-2.0176	0.0566	0.0441	-0.2255	-0.0030
large	0.1868	3.2083	0.0582	0.0014	0.0724	0.3012
largest	0.4918	5.0989	0.0964	0.0000	0.3023	0.6812
powder_room	0.1536	2.2284	0.0689	0.0263	0.0182	0.2891
fair	-0.2550	-5.8950	0.0432	0.0000	-0.3399	-0.1700
poor	-0.3630	-3.4457	0.1053	0.0006	-0.5699	-0.1561
yb_post_1961	0.4703	4.1870	0.1123	0.0000	0.2497	0.6910
crawl	-0.1556	-2.2642	0.0687	0.0240	-0.2905	-0.0206
ln_garagespaces	0.0800	1.7622	0.0454	0.0786	-0.0092	0.1692
elec_baseboard	NA	NA	NA	NA	NA	NA
ecf_6R602	0.1786	1.8595	0.0960	0.0635	-0.0101	0.3672
ecf_6R604	0.4281	5.2720	0.0812	0.0000	0.2686	0.5876
ecf_6R605	0.6462	6.9073	0.0935	0.0000	0.4624	0.8299
ecf_6R606	0.3037	2.2683	0.1339	0.0237	0.0407	0.5667
ecf_6R607	0.9517	6.8446	0.1390	0.0000	0.6785	1.2248
ecf_6R608	1.9464	10.4690	0.1859	0.0000	1.5812	2.3116
ecf_6R609	2.0559	4.2151	0.4878	0.0000	1.0978	3.0141
ecf_6R611	0.7646	9.9705	0.0767	0.0000	0.6139	0.9152
ecf_6R612	0.7698	6.1763	0.1246	0.0000	0.5250	1.0146
ecf_6R613	1.0096	9.5306	0.1059	0.0000	0.8015	1.2177
ecf_6R614	2.0876	9.0781	0.2300	0.0000	1.6359	2.5393
ecf_6R615	0.8544	8.6397	0.0989	0.0000	0.6601	1.0486
ecf_6R616	1.2440	5.1623	0.2410	0.0000	0.7707	1.7174
ecf_6R617	1.2168	4.3988	0.2766	0.0000	0.6734	1.7602
ecf_6R618	0.3417	2.1532	0.1587	0.0317	0.0300	0.6535
ecf_6R620	0.4414	5.8050	0.0760	0.0000	0.2921	0.5908
months_1to9	0.0186	1.9027	0.0098	0.0576	-0.0006	0.0377
months_10to16	-0.0207	-1.8292	0.0113	0.0679	-0.0430	0.0015
months_17to24	0.0682	4.9263	0.0138	0.0000	0.0410	0.0954

Residuals for LN_PRICE (Inliers) Model

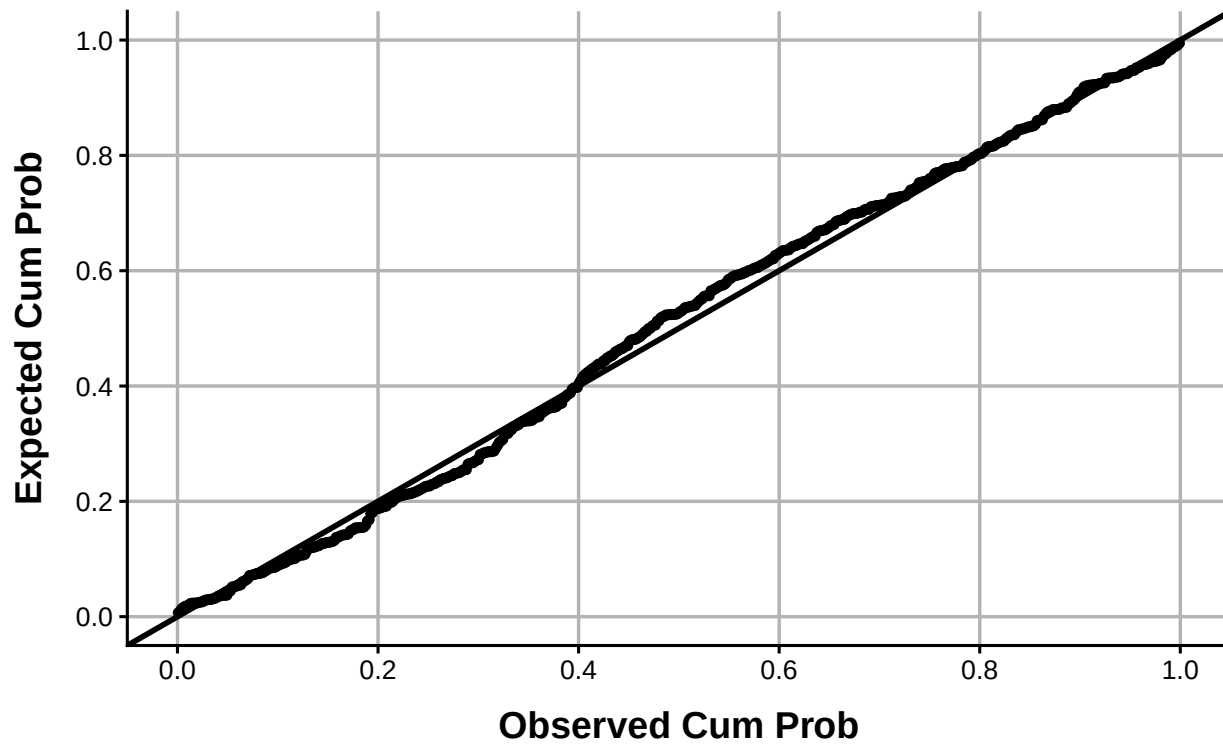


Histogram

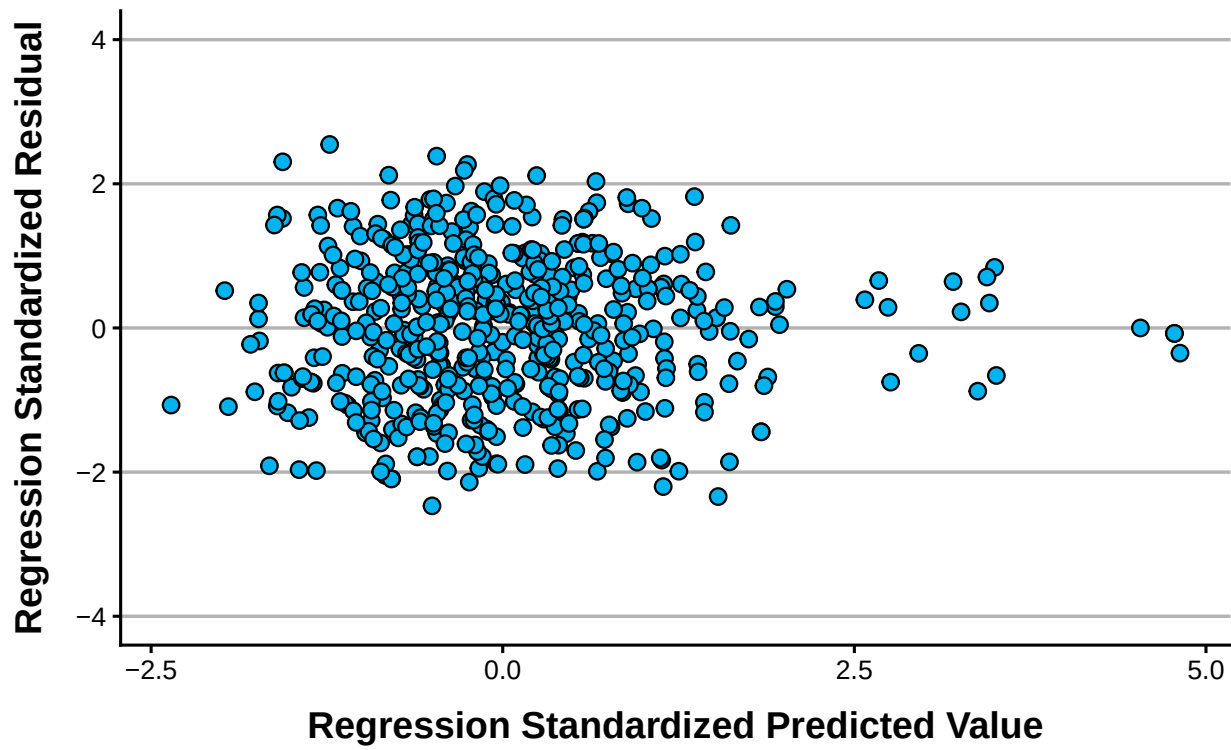
Dependent Variable: LN\ PRICE



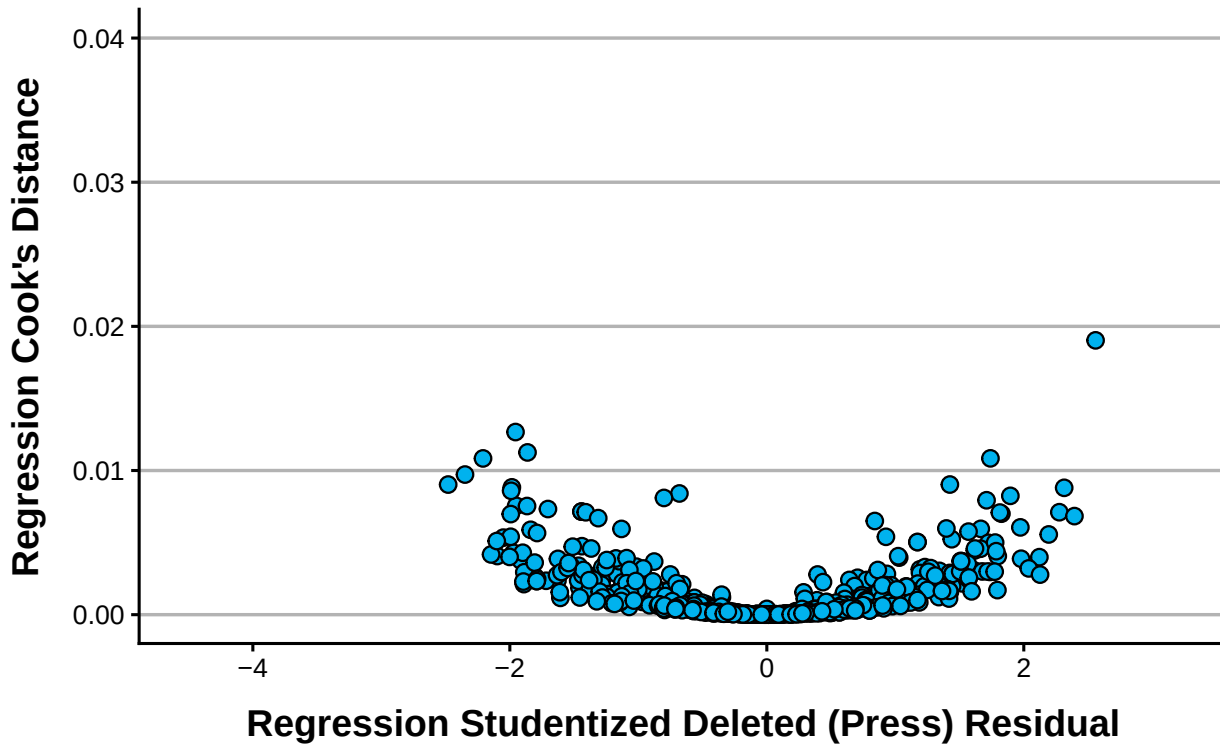
Normal P-P Plot of Regression Standardized Residual
Dependent Variable: LN_PRICE



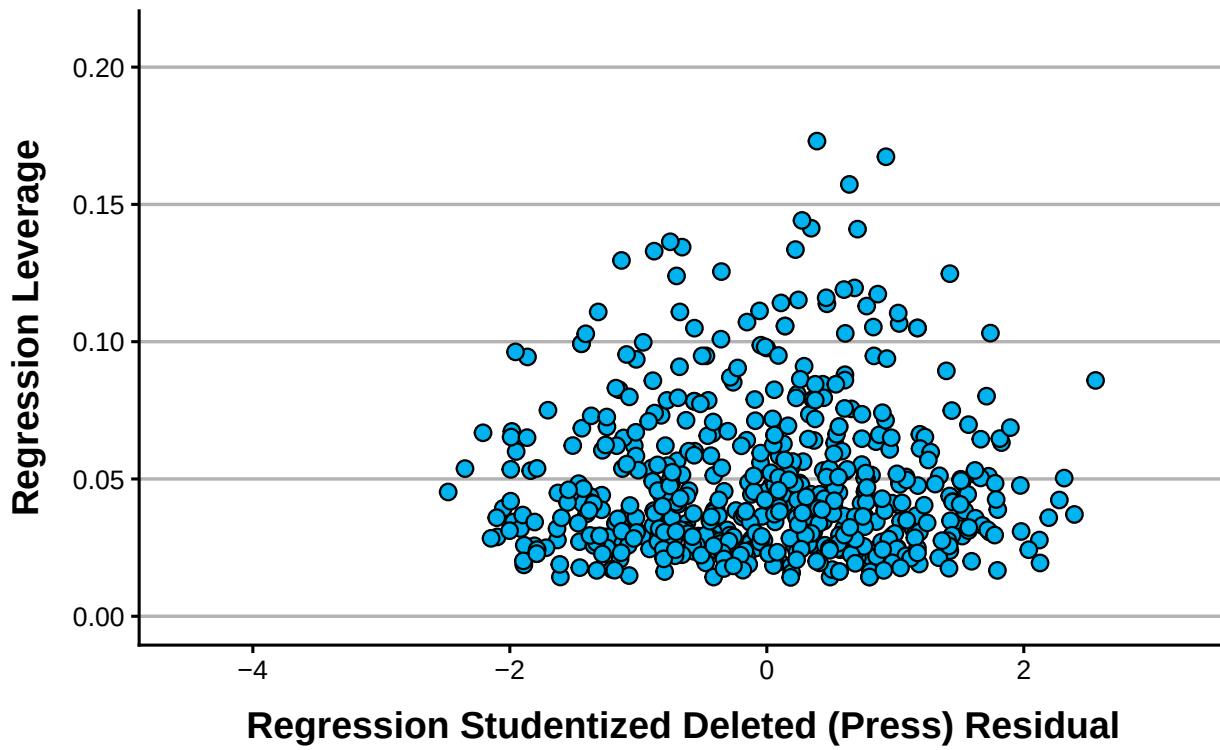
Scatterplot
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE



Scatterplot
Dependent Variable: LN_PRICE



Model IV

Table 10: District 6 LN_TASP Model I assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
622	1.361257	1.5936	1.264782	1.259981	46.86939	-0.7445096

Table 11: District 6 LN_TASP Model II assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
622	1.357444	1.593452	1.266206	1.258446	46.94645	-0.7454256

Table 12: District 6 LN_TASP Model I fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
578	547	33	0.4584	0.6042	0.5825	0.4459	0.3676	27.8331	0	770.6944	910.2007

Table 13: District 6 LN_TASP Model I coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.5880	114.3768	0.0926	0.0000	10.4061	10.7698
arterial	0.2514	3.5129	0.0716	0.0005	0.1108	0.3920
single_family1story	0.1284	2.5022	0.0513	0.0126	0.0276	0.2292
half_duplex1story	NA	NA	NA	NA	NA	NA
smallest	-0.3149	-4.1797	0.0753	0.0000	-0.4628	-0.1669
small	-0.1122	-1.9840	0.0565	0.0478	-0.2232	-0.0011
large	0.1860	3.1965	0.0582	0.0015	0.0717	0.3004
largest	0.4866	5.0608	0.0961	0.0000	0.2977	0.6754
powder_room	0.1536	2.2292	0.0689	0.0262	0.0183	0.2890
fair	-0.2522	-5.8556	0.0431	0.0000	-0.3368	-0.1676
poor	-0.3607	-3.4268	0.1053	0.0007	-0.5675	-0.1539
yb_post_1961	0.4659	4.1550	0.1121	0.0000	0.2456	0.6861
crawl	-0.1499	-2.1962	0.0682	0.0285	-0.2839	-0.0158
ln_garagespaces	0.0822	1.8160	0.0453	0.0699	-0.0067	0.1712
elec_baseboard	NA	NA	NA	NA	NA	NA
ecf_6R602	0.1807	1.8835	0.0959	0.0602	-0.0078	0.3691
ecf_6R604	0.4323	5.3401	0.0810	0.0000	0.2733	0.5914
ecf_6R605	0.6552	7.0665	0.0927	0.0000	0.4730	0.8373
ecf_6R606	0.2983	2.2321	0.1336	0.0260	0.0358	0.5608
ecf_6R607	0.9475	6.8231	0.1389	0.0000	0.6747	1.2203
ecf_6R608	1.9480	10.4827	0.1858	0.0000	1.5829	2.3130
ecf_6R609	2.0561	4.2171	0.4876	0.0000	1.0984	3.0138
ecf_6R611	0.7647	9.9766	0.0767	0.0000	0.6142	0.9153
ecf_6R612	0.7684	6.1684	0.1246	0.0000	0.5237	1.0131
ecf_6R613	1.0083	9.5237	0.1059	0.0000	0.8003	1.2163
ecf_6R614	2.0951	9.1236	0.2296	0.0000	1.6440	2.5462
ecf_6R615	0.8561	8.6633	0.0988	0.0000	0.6620	1.0502
ecf_6R616	1.2538	5.2132	0.2405	0.0000	0.7814	1.7263
ecf_6R617	1.2165	4.3997	0.2765	0.0000	0.6734	1.7597
ecf_6R618	0.3386	2.1351	0.1586	0.0332	0.0271	0.6501
ecf_6R620	0.4423	5.8201	0.0760	0.0000	0.2930	0.5916
months_1to9	0.0176	1.8224	0.0097	0.0689	-0.0014	0.0366
months_10to16	-0.0155	-1.7535	0.0089	0.0801	-0.0329	0.0019

Table 14: District 6 LN_TASP Model II fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
578	548	32	0.4585	0.6032	0.5822	0.4465	0.3697	28.7264	0	770.1384	905.2852

Table 15: District 6 LN_TASP Model II coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.6489	135.5674	0.0786	0.0000	10.4946	10.8032
arterial	0.2453	3.4317	0.0715	0.0006	0.1049	0.3856
single_family1story	0.1342	2.7361	0.0490	0.0064	0.0379	0.2305
half_duplex1story	NA	NA	NA	NA	NA	NA
smallest	-0.3081	-4.0986	0.0752	0.0000	-0.4558	-0.1604
small	-0.1152	-2.0429	0.0564	0.0415	-0.2259	-0.0044
large	0.1819	3.1220	0.0583	0.0019	0.0675	0.2964
largest	0.4813	4.9999	0.0963	0.0000	0.2922	0.6704
powder_room	0.1650	2.3890	0.0690	0.0172	0.0293	0.3006
good	0.2601	1.7621	0.1476	0.0786	-0.0299	0.5500
fair	-0.2411	-5.5594	0.0434	0.0000	-0.3264	-0.1559
poor	-0.3537	-3.3551	0.1054	0.0008	-0.5607	-0.1466
yb_post_1961	0.4524	4.0300	0.1123	0.0001	0.2319	0.6730
crawl	-0.1420	-2.0942	0.0678	0.0367	-0.2753	-0.0088
garage	0.0701	1.7652	0.0397	0.0781	-0.0079	0.1481
elec_baseboard	NA	NA	NA	NA	NA	NA
ecf_6R602	0.1810	1.8891	0.0958	0.0594	-0.0072	0.3692
ecf_6R604	0.4409	5.4516	0.0809	0.0000	0.2820	0.5997
ecf_6R605	0.6480	7.0262	0.0922	0.0000	0.4668	0.8291
ecf_6R606	0.3067	2.2921	0.1338	0.0223	0.0439	0.5696
ecf_6R607	0.9376	6.7662	0.1386	0.0000	0.6654	1.2098
ecf_6R608	1.9558	10.5088	0.1861	0.0000	1.5902	2.3214
ecf_6R609	2.1666	4.4598	0.4858	0.0000	1.2123	3.1209
ecf_6R611	0.7578	9.9245	0.0764	0.0000	0.6078	0.9078
ecf_6R612	0.7606	6.1682	0.1233	0.0000	0.5184	1.0028
ecf_6R613	1.0167	9.5732	0.1062	0.0000	0.8081	1.2253
ecf_6R614	1.8909	7.5141	0.2516	0.0000	1.3966	2.3852
ecf_6R615	0.8335	8.4197	0.0990	0.0000	0.6391	1.0280
ecf_6R616	1.2450	5.1860	0.2401	0.0000	0.7734	1.7166
ecf_6R617	1.1303	4.0456	0.2794	0.0001	0.5815	1.6791
ecf_6R618	0.3472	2.1969	0.1580	0.0284	0.0368	0.6576
ecf_6R620	0.4411	5.8651	0.0752	0.0000	0.2934	0.5888

